Chapter 11. Mediation, SWIGs and G-Methods

Direct and indirect effects, single-world intervention graphs, the g-formula and its estimators

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11.1 Direct and indirect effects

Motivation. How to split a total effect into direct and mediated parts in nonlinear models, where "hold the mediator fixed and subtract" no longer defines an indirect effect.

11.1.1 Preliminaries: potential responses

Motivation. Counterfactual notation, including nested counterfactuals.

In an SCM, $Y(u)$ is the value of Y for unit $U = u$; $Y_x(u)$ its value under $\mathrm{do}(X = x)$, so $P(Y_x = y) = P(y \mid \mathrm{do}(x))$; $Y_{xz}(u)$ its value under $\mathrm{do}(X = x, Z = z)$; $Z_{x^*}(u)$ the mediator value under $\mathrm{do}(X = x^*)$. The nested counterfactual

$$Y_{x,Z_{x^*}}(u): \quad \text{set } X = x, \text{ but set } Z \text{ to the value it would have taken under } X = x^*.$$

11.1.2 Introduction

Motivation. Direct and indirect effects, and the prescriptive/descriptive distinction.

For $X \to Z \to Y$ with $X \to Y$:

- **Total effect:** the change in the distribution of Y under $\mathrm{do}(X = x)$, $P(Y_x = y) = P(y \mid \mathrm{do}(x))$.
- **Direct effect:** measured holding all intermediate variables constant.
- **Indirect effect:** in a linear system, total − direct; in a nonlinear system not measurable by holding variables constant.

Holding the other factors fixed cuts every path from X to Y except $X \to Y$, which defines the direct effect. No choice of variables to hold constant isolates the mediated paths, so indirect effects need another definition.

11.1.2.1 Examples

Birth-control pill and thrombosis. The pill may raise thrombosis directly but lower it by preventing pregnancy. The direct effect is biologically stable and more transportable than the total effect.

Race/sex discrimination in hiring. The legal question is the direct effect: same decision had race or gender differed, all else equal?

Smoking, hypertension and heart disease (Hernán and Robins 2020, Ch. 23). Quitting smoking ($A = 0$) reduces myocardial infarction Y; direct: $A \to Y$, indirect: $A \to M \to Y$ via hypertension M. If a drug removed smoking-induced hypertension, how much benefit would remain?

Drug, aspirin and headache. A drug causes headaches, so patients take aspirin, which may affect the disease or modify the drug's effect. A direct effect

$$P(Y_{xz} = y) - P(Y_{x^*z} = y),$$

with z a specified aspirin level, isolates the drug. In linear systems it equals a path coefficient for every z; in nonlinear systems it depends on z, which must represent the target policy. Hence two conceptualisations.

11.1.2.2 Prescriptive versus descriptive

- **Prescriptive (controlled direct effect):** would an untreated patient improve if treated, with aspirin held at a set level $Z = z$? No individual behavior needed: $P(Y_{xz} = y) - P(Y_{x^*z} = y)$.
- **Descriptive (natural direct effect):** would the patient improve if treated, with aspirin at the level the patient takes untreated? Requires individual natural behavior.

If a patient takes aspirin iff treated and treatment works only with aspirin, the NDE is zero but the CDE (at $z =$ aspirin) is not. The prescriptive form is *pragmatic* (a dose prescribed to all), the descriptive *attributional* (drug or aspirin use?).

Why the average natural direct effect? Would the drug stay effective without the headache side effect? The CDE fixes one aspirin level, but aspirin use varies for other reasons that may modulate effectiveness; the average NDE answers this.

Descriptive indirect effects. Only the descriptive definition transfers: untreated, would the patient recover given the aspirin taken under treatment? E.g. if asking about gender is banned, residual sex preference in hiring is the indirect effect via qualifications, predicted from pre-policy data as an average indirect effect.

11.1.3 Mathematisation

Motivation. Controlled and natural direct effects, and identification of the latter.

X is the control variable, Y the response, Z the set of intermediate variables, $Y_x(u)$ as above.

Definition (Controlled direct effect)

$\text{CDE}_z(x, x^*; Y, u) = Y_{xz}(u) - Y_{x^*z}(u)$, and $\text{CDE}_z(x, x^*; Y) = \mathbb{E}[Y_{xz} - Y_{x^*z}]$, the expectation taken over u.

(Other contrasts, e.g. ratios, can replace the difference.)

Definition (Natural direct effect)

$\text{NDE}(x, x^*; Y, u) = Y_{x,Z_{x^*}(u)}(u) - Y_{x^*}(u)$, and $\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$.

11.1.3.1 The cross-world problem

$Y_{x,Z_{x^*}}$ involves two worlds, $X = x$ (for Y) and $X = x^*$ (for Z_{x^*}), never observed together, even with joint randomization of X and Z. NDE and NIE are unverifiable in principle and need the untestable *cross-world independence* $Y_{xz} \perp\!\!\!\perp Z_{x^*}$ (Hernán and Robins 2020, §23.1; Robins and Richardson 2010).

11.1.3.2 Experimental identification

$\text{NDE}(x, x^*; Y) = \mathbb{E}[Y_{x,Z_{x^*}}] - \mathbb{E}[Y_{x^*}]$ is in general not reducible to terms $P(Y_x = y)$ or $P(Y_{xz} = y)$. (General nested-counterfactual identification came later: Shpitser and Pearl 2008; Correa, Lee and Bareinboim 2021.)

Theorem 11.1 (Pearl 2001)

If there is a set W of covariates, non-descendants of X or Z, with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$, the average NDE is experimentally identifiable:

$$\text{NDE}(x, x^*; Y) = \sum_{w,z} \big[\mathbb{E}(Y_{xz} \mid w) - \mathbb{E}(Y_{x^*z} \mid w)\big] P(Z_{x^*} = z \mid w) P(w).$$

Proof [Supplement].

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid Z_{x^*} = z, w) P(Z_{x^*} = z \mid w) P(w) = \sum_w \sum_z \mathbb{E}(Y_{xz} \mid w) P(Z_{x^*} = z \mid w) P(w),$$
$$\mathbb{E}[Y_{x^*}] = \mathbb{E}[Y_{x^*,Z_{x^*}}] = \sum_w \sum_z \mathbb{E}(Y_{x^*z} \mid w) P(Z_{x^*} = z \mid w) P(w),$$

using $Y_{xz} = Y_{x,Z_{x^*}}$ on $\{Z_{x^*} = z\}$ and the conditional independence (first line), and composition $Y_{x^*} = Y_{x^*,Z_{x^*}}$ (second). Subtract. ■

Graphical criterion. (a): $X \to Z \to Y$, $X \to Y$, W a common cause of Z and Y. (b): in $\mathcal{G}_{\underline{XZ}}$ (arrows out of X and Z deleted), $(Y \perp_d Z \mid W)_{\mathcal{G}_{\underline{XZ}}}$, which ensures the condition.

11.1.3.3 Markovian models

The condition holds with $W = \emptyset$ when Y_{xz} is independent of all model variables. In Markovian models:

- $P(Y_{xz} = y) = P(y \mid x, z)$, where $X \cup Z$ is the parent set of Y;
- $P(Z_{x^*} = z) = \sum_s P(z \mid x^*, s) P(s)$;
- $P(Y_{x,Z_{x^*}} = y) = \sum_s \sum_z P(y \mid x, z) P(z \mid x^*, s) P(s)$,

with S the parents of Z other than X (or any back-door set). Hence, from non-experimental data,

$$\text{NDE}(x, x^*; Y) = \sum_s \sum_z \big[\mathbb{E}(Y \mid x, z) - \mathbb{E}(Y \mid x^*, z)\big] P(z \mid x^*, s) P(s).$$

11.1.3.4 The mediation formula

For $X \to Z \to Y$, $X \to Y$, with no Z–Y confounding, Y under x with the mediator distribution of x^*:

$$\mathbb{E}[Y_{x,Z_{x^*}}] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]\, P(Z = z \mid X = x^*).$$

Smoking example: MI risk if patients kept smoking ($a = 1$) with the blood pressure of quitters ($a = 0$):

$$\mathbb{E}[Y_{1,M_0}] = \sum_m \mathbb{E}[Y \mid A = 1, M = m]\, P(M = m \mid A = 0).$$

11.1.4 Natural indirect effects

Motivation. The NIE, its relation to the total effect, and its identification.

Definition (Natural indirect effect)

$\text{NIE}(x, x^*; Y, u) = Y_{x^*,Z_x(u)}(u) - Y_{x^*}(u)$, and $\text{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*})$: treatment at the reference level, mediator as under treatment.

Decomposition.

$$\begin{aligned}\mathrm{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}),\\ \mathrm{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x, Z_{x^*}}) - \mathbb{E}(Y_{x^*}),\\ -\mathrm{NIE}(x^*, x; Y) &= -\mathbb{E}(Y_{x, Z_{x^*}}) + \mathbb{E}(Y_x).\end{aligned}$$

Adding the last two lines,

$$\mathrm{TE}(x, x^*; Y) = \mathrm{NDE}(x, x^*; Y) - \mathrm{NIE}(x^*, x; Y) :$$

the direct effect of $x^* \to x$ minus the indirect effect of the reverse transition. In the linear case this is $\mathrm{TE} = \mathrm{NDE}(x, x^*; Y) + \mathrm{NIE}(x, x^*; Y)$ (Appendix 11.1.6).

Theorem 11.2 (Identification of the NIE)

If there is a set W of non-descendants of X or Z with $Y_{xz} \perp\!\!\!\perp Z_{x^*} \mid W$ for all x and z, then

$$\mathrm{NIE}(x, x^*; Y) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)\big[P(Z_x = z \mid w) - P(Z_{x^*} = z \mid w)\big]P(w).$$

In non-experimental studies the NIE is identifiable when $\mathbb{E}(Y_{x^* z} \mid w)$, $P(Z_x = z \mid w)$ and $P(Z_{x^*} = z \mid w)$ are. In a simple Markovian model,

$$\mathrm{NIE}(x, x^*; Y) = \sum_{z} \mathbb{E}(Y \mid x^*, z)\big[P(z \mid x) - P(z \mid x^*)\big].$$

Proof [Supplement]. By composition, $\mathrm{NIE}(x, x^*; Y) = \mathbb{E}(Y_{x^*, Z_x}) - \mathbb{E}(Y_{x^*, Z_{x^*}})$. Conditioning on W and $Z_x = z$ and using $Y_{x^* z} \perp\!\!\!\perp Z_x \mid W$,

$$\mathbb{E}(Y_{x^*, Z_x}) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid Z_x = z, w)P(Z_x = z \mid w)P(w) = \sum_{w,z} \mathbb{E}(Y_{x^* z} \mid w)P(Z_x = z \mid w)P(w).$$

The same with x^* for x gives the second term; subtract. In a Markovian model with $W = \emptyset$, $\mathbb{E}(Y_{x^* z}) = \mathbb{E}(Y \mid x^*, z)$ and $P(Z_x = z) = P(z \mid x)$. ■

11.1.5 Path-specific effects

Motivation. Effects along any chosen set of paths.

With several mediators ($X \to W \to Z \to Y$, $X \to Z$, $X \to Y$, ...), the effect along selected paths is obtained by feeding each mechanism its inputs under x or under x^*, according to whether the edge is selected.

Definition (Path-specific effect; Pearl 2001)

Let $\mathcal{G}$ be the causal graph of $\mathcal{M}$ and g an edge-subgraph of $\mathcal{G}$ containing the paths selected for analysis. The g-specific effect of x on Y (relative to the reference x^*) is the total effect of x on Y in a modified model $\mathcal{M}_g$ built as follows. Partition each parent set as $Pa_i = \{Pa_i(g), Pa_i(\bar{g})\}$, where $Pa_i(g)$ are the parents linked to X_i in g and $Pa_i(\bar{g})$ the rest. Replace each $f_i(\mathbf{pa}_i, u)$ by

$$f_i(\mathbf{pa}_i, u; g) = f_i(\mathbf{pa}_i(g), \mathbf{pa}_i^*(\bar{g}), u),$$

where $\mathbf{pa}_i^*(\bar{g}) = Pa_i(\bar{g})_{x^*}$ are the values those variables would attain (in $\mathcal{M}$ and u) under $X = x^*$. Then $SE_g(x, x^*; Y, u)_{\mathcal{M}} = TE(x, x^*; Y, u)_{\mathcal{M}_g}$.

$g = \{X \to Y\}$ gives the NDE; g the mediated paths, the indirect effect. This leads to nested counterfactuals like $Y_{W_x, Z_x, Z_{x^*}}$, represented in counterfactual graphs.

Conclusions. Path switching, rather than variable fixing, extends direct and indirect effects to nonlinear models and gives indirect effects an operational meaning. They can be identifiable even when Z is affected by X, and always are in Markovian models.

11.1.6 Appendix: The linear additive relation

Let $Z = \alpha X + U_Z$, $Y = \beta X + \gamma Z + U_Y$, so $Z_x = \alpha x + U_Z$, $Z_{x^*} = \alpha x^* + U_Z$. Then

$$\begin{aligned}\text{TE}(x, x^*; Y) &= \mathbb{E}(Y_x) - \mathbb{E}(Y_{x^*}) = (\beta + \alpha\gamma)(x - x^*),\\ \text{NDE}(x, x^*; Y) &= \mathbb{E}(Y_{x,Z_{x^*}}) - \mathbb{E}(Y_{x^*}) = \beta(x - x^*),\\ \text{NIE}(x, x^*; Y) &= \mathbb{E}(Y_{x^*,Z_x}) - \mathbb{E}(Y_{x^*}) = \alpha\gamma(x - x^*),\end{aligned}$$

so TE = NDE + NIE; also $\text{NIE}(x^*, x; Y) = -\alpha\gamma(x - x^*)$, consistent with $\text{TE} = \text{NDE}(x, x^*) - \text{NIE}(x^*, x)$. Nonlinear models need not preserve this sum.

11.1.7 Appendix: Counterfactual axioms

Composition. For singleton Y, W and any X: $W_x(u) = w \Rightarrow Y_{xw}(u) = Y_x(u)$ (forcing a variable to its natural value changes nothing).
Consistency. $X(u) = x \Rightarrow Y(u) = Y_x(u)$.
Effectiveness. For all X, W: $X_{xw}(u) = x$.
Reversibility. For Y, W and any X: $(Y_{xw}(u) = y) \wedge (W_{xy}(u) = w) \Rightarrow Y_x(u) = y$.

11.2 Single-world intervention graphs

Motivation. How to put potential outcomes on a causal graph and read counterfactual independences by d-separation, without the extra untestable assumptions of a full structural-equation model.

11.2.1 Potential outcomes and graphs

Motivation. The two languages that SWIGs unify.

11.2.1.1 Potential outcomes

For binary X and Y: $Y(x = 0)$ (or $Y(x_0)$) is the value of Y for a unit assigned $X = 0$; $Y(x = 1)$ (or $Y(x_1)$) likewise. This presumes one version of treatment and no interference (SUTVA). "Potential outcome" and "counterfactual" are synonyms.

Response types for binary Y:

$Y(x_0)$	$Y(x_1)$	Name
0	0	never recover
0	1	helped
1	0	hurt
1	1	always recover

Relating counterfactuals and do-notation. $P(Y(x) = y) \equiv P(Y = y \mid \text{do}(X = x))$, but counterfactuals say more, e.g. $P(Y(x = 0) = y \mid X = 1)$, outcomes of the treated had they not been treated. Under randomization $X \perp\!\!\!\perp Y(x = 0)$ and it equals $P(Y(x = 0) = y)$; observationally it may not.

11.2.1.2 Graphs

A DAG carries the factorization $P(\mathbf{V}) = \prod_{X \in \mathbf{V}} P(X \mid \text{pa}(X))$; e.g. for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$, $P(A,B,C,D,E) = P(A)P(B \mid A)P(C \mid A)P(D \mid B,C)P(E \mid D)$, and d-separation gives $C \perp\!\!\!\perp B \mid A$, $D \perp\!\!\!\perp A \mid B,C$, $E \perp\!\!\!\perp A,B,C \mid D$.

Edges are direct causal relations, and confounders (e.g. unobserved H with $X \leftarrow H \to Y$) are included (Wright's path diagrams, 1920s). For unconfounded $X \to Y$, $P(x,y) = P(x)P(y \mid x)$ and

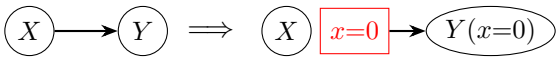
$$P(Y(x) = y) = P(Y = y \mid \text{do}(X = x)) = P(Y = y \mid X = x),$$

$P(y \mid \text{do}(x))$ being the distribution in an experiment fixing $X = x$. No confounding means $X \perp\!\!\!\perp Y(x_0)$ and $X \perp\!\!\!\perp Y(x_1)$; confounding, $X \not\perp\!\!\!\perp Y(x_0)$ or $X \not\perp\!\!\!\perp Y(x_1)$. But $Y(x_0)$, $Y(x_1)$ do not appear on these graphs.

11.2.2 Node splitting

Motivation. Put counterfactual variables on the graph.

Setting X to 0. From $P(X = \tilde{x}, Y = \tilde{y}) = P(X = \tilde{x})P(Y = \tilde{y} \mid X = \tilde{x})$ for $X \to Y$, split X into a random node X and a fixed node $x = 0$; the outcome becomes $Y(x = 0)$:



$X \perp\!\!\!\perp Y(x = 0)$ is now read off, with factorization

$$P(X = \tilde{x}, Y(x = 0) = \tilde{y}) = P(X = \tilde{x})\, P(Y(x = 0) = \tilde{y}), \qquad P(Y(x = 0) = \tilde{y}) = P(Y = \tilde{y} \mid X = 0).$$

The last equation, linking the two factorizations, is the *modularity assumption*.

Marginals represented by SWIGs are identified. $\mathcal{G}(x_0)$ represents $P(X, Y(x_0))$ and $\mathcal{G}(x_1)$ represents $P(X, Y(x_1))$, both identified from $P(X,Y)$ without confounding; $P(X, Y(x_0), Y(x_1))$ is not. $Y(x = 0)$ and $Y(x = 1)$ never share a graph: we assume $X \perp\!\!\!\perp Y(x = 0)$ and $X \perp\!\!\!\perp Y(x = 1)$ but not $X \perp\!\!\!\perp \{Y(x = 0), Y(x = 1)\}$. Hence *single-world* intervention graphs.

Templates. The template $\mathcal{G}(x)$: $X \mid x \to Y(x)$ is a graph-valued function returning a SWIG $\mathcal{G}(x)$ for each x, each a different margin.

Intuition (Robins, VanderWeele and Richardson 2007). Whenever a patient swallows the drug ($X = 1$), instantly give a side-effect-free emetic that expels it before absorption; the recorded outcome is $Y(x = 0)$, observed jointly with the natural choice X.

11.2.3 SWIG construction and properties

Motivation. General SWIG construction and its implications.

Given $\mathcal{G}$ and $\mathbf{A} = \{A_1, \ldots, A_k\}$, form $\mathcal{G}(\mathbf{a})$:

1. **Node splitting:** split each $A \in \mathbf{A}$ into a random node A, inheriting edges *into* A, and a fixed node a, inheriting edges *out of* A.
2. **Relabelling:** each descendant Y of a fixed node becomes $Y(\mathbf{a}_Y)$, indexed by its fixed-node ancestors (e.g. $B(a, \ldots)$, $C(a, \ldots)$, $D(a, \ldots)$); non-descendants keep their names.

Factorization and modularity. $\mathcal{G}$ carries the observed $P(\mathbf{V})$; $\mathcal{G}(\tilde{\mathbf{a}})$ carries $P(\mathbf{V}(\tilde{\mathbf{a}}))$.

- *Factorization:* $P(\mathbf{V}(\tilde{\mathbf{a}}))$ factorizes with respect to $\mathcal{G}(\tilde{\mathbf{a}})$, ignoring fixed nodes.

- *Modularity:* the conditional density of $Y(\tilde{\mathbf{a}}_Y)$ in $\mathcal{G}(\tilde{\mathbf{a}})$ equals that of Y in $\mathcal{G}$ with $\tilde{a}_i$ substituted for each parent $A_i \in \mathbf{A}$ of Y.

Hence $P(\mathbf{V}(\tilde{\mathbf{a}}))$ is identified from $P(\mathbf{V})$.

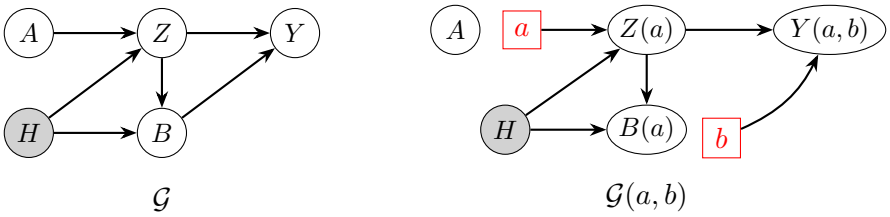
d-separation on SWIGs. If random-node sets $\mathbf{B}(\tilde{\mathbf{a}})$, $\mathbf{C}(\tilde{\mathbf{a}})$ are d-separated by $\mathbf{D}(\tilde{\mathbf{a}})$ and the fixed nodes $\tilde{\mathbf{a}}$ in $\mathcal{G}(\tilde{\mathbf{a}})$, then

$$\mathbf{B}(\tilde{\mathbf{a}}) \perp\!\!\!\perp \mathbf{C}(\tilde{\mathbf{a}}) \mid \mathbf{D}(\tilde{\mathbf{a}}) \quad \text{in } P(\mathbf{V}(\tilde{\mathbf{a}})).$$

11.2.4 A harder inferential problem

Motivation. A SWIG settles a counterfactual independence query.

$\mathcal{G}$: $A \to Z$, $Z \to Y$, $Z \to B$, $B \to Y$, unobserved $H \to Z$, $H \to B$. Does $\mathcal{G}$ imply $Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A$?



Pearl (2009, Ex. 11.3.3) claims the DAG does not imply

$$Y(a,b) \perp\!\!\!\perp B \mid Z, A = a. \tag{3}$$

The SWIG shows (3) holds. In $\mathcal{G}(a,b)$ every path from $B(a)$ to $Y(a,b)$ passes through the non-collider $Z(a)$ ($B(a) \leftarrow Z(a) \to Y(a,b)$, $B(a) \leftarrow H \to Z(a) \to Y(a,b)$), and $B \to Y$ now leaves the fixed node b. Hence

$$Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A \quad \Longrightarrow \quad Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A = a,$$

equivalent to (3) by consistency ($B(a) = B$, $Z(a) = Z$ when $A = a$). (Pearl had used the failure of (3) to reject a claim of Robins.)

11.2.5 Adjustment for confounding

Motivation. The back-door formula in SWIG language.

Graph $L \to X$, $L \to Y$, $X \to Y$; SWIG $\mathcal{G}(x)$: $L \to X$, $L \to Y(x)$, $x \to Y(x)$, so $X \perp\!\!\!\perp Y(x) \mid L$. Then

$$\begin{aligned} P[Y(\tilde{x}) = y] &= \sum_l P[Y(\tilde{x}) = y \mid L = l] P(L = l) \\ &= \sum_l P[Y(\tilde{x}) = y \mid L = l, X = \tilde{x}] P(L = l) && (X \perp\!\!\!\perp Y(\tilde{x}) \mid L) \\ &= \sum_l P[Y = y \mid L = l, X = \tilde{x}] P(L = l) && \text{(consistency)}. \end{aligned}$$

Connection to do-calculus. Factorization and modularity imply all identification results of the do-calculus (Pearl 1995; Spirtes et al. 1993): $P(Y = y \mid \mathrm{do}(A = a))$ is identified iff $P(Y(a) = y)$ is.

11.2.6 Conclusion

Eliminating a false trichotomy. Before SWIGs, the options were graphs without counterfactuals (Dawid), counterfactuals without graphs, or Pearl's NPSEMs with many assumptions untestable even in principle. SWIGs combine both without them.

Summary and extensions. Node splitting links the factorizations of the DAG and the SWIG, whose counterfactual distribution is identified from the observed one. (Not covered: individual- vs. population-level absence of direct effects; partially well-defined interventions.)

References: Pearl (2009); Richardson and Robins (2013), CSSS Technical Report 128; Robins (1986); Robins, VanderWeele and Richardson (2007); Shpitser and Pearl (2008); Spirtes, Glymour and Scheines (1993).

11.3 G-methods for time-varying treatments

Motivation. How to estimate the effect of a treatment sequence when covariates are both affected by earlier treatment and drive later treatment, where regression, stratification and matching fail.

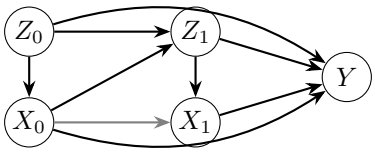
11.3.1 Time-varying treatments

Motivation. Sequential decisions, and why feedback breaks standard adjustment.

Recall the single time point. Back-door adjustment $\mathbb{E}[Y(x)] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]P(Z = z)$; IPW $\mathbb{E}[Y(x)] = \mathbb{E}[\mathbb{1}(X = x)Y/P(X = x \mid Z)]$.

With repeated decisions (medication at each visit, staged policies, adaptive algorithms), the process is history-dependent: action $\to$ state $\to$ action $\to \cdots \to$ outcome.

Causal diagrams. Treatments X_k, history $\bar{X}_k = (X_0, \ldots, X_k)$; covariates Z_k, history $\bar{Z}_k = (Z_0, \ldots, Z_k)$; final outcome Y. Two periods: $Z_0 \to X_0$, $Z_0 \to Z_1$, $X_0 \to Z_1$, $Z_1 \to X_1$, $X_0 \to X_1$, all $\to Y$.



Treatment–confounder feedback: (1) X_{k-1} affects confounder Z_k and (2) Z_k affects X_k. Example (HIV): ART $(X) \to$ CD4 count or viral load $(Z) \to$ next ART decision $\to \cdots$, possibly also $X_{k-1} \to X_k$.

Treatment strategies. *Static*: the same for everyone, e.g. "always treat" $\bar{x} = (1, \ldots, 1)$, "never treat" $\bar{x} = (0, \ldots, 0)$. *Dynamic*: $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, e.g. "start ART when CD4 drops below 500". Estimands: $\mathbb{E}[Y(\bar{x})] - \mathbb{E}[Y(\bar{x}')]$.

11.3.1.1 Identification assumptions

In a **sequentially randomized experiment** X_k is randomized with probability depending only on $(\bar{X}_{k-1}, \bar{Z}_k)$; observationally, no unmeasured common causes of X_k and Y given $(\bar{X}_{k-1}, \bar{Z}_k)$. An unmeasured U may affect Y and Z_1 if it has no arrow into X_0 or X_1.

- **Sequential exchangeability:** $Y(\bar{x}) \perp\!\!\!\perp X_k \mid \bar{X}_{k-1}, \bar{Z}_k$, $k = 0, \ldots, K$.

- **Consistency:** $Y(\bar{x}) = Y$ when $\bar{X} = \bar{x}$.
- **Positivity:** $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k) > 0$.

11.3.1.2 The fundamental dilemma

Adjusting for Z_1 removes confounding of $X_1 \to Y$ but blocks $X_0 \to Z_1 \to Y$ (and, with $U \to Z_1$, $U \to Y$, opens the collider path $X_0 \to Z_1 \leftarrow U \to Y$). Not adjusting leaves X_1 confounded. Regression, stratification and matching cannot do both.

G-methods (Robins) are consistent under time-varying confounding:

(1) the g-formula (outcome modeling);

(2) IP-weighted marginal structural models (treatment modeling);

(3) g-estimation of structural nested models (effect modeling). All share the same assumptions and handle Z_1 without blocking the path through it.

What you already know. The truncated factorization (Chapter 2) for (Z_0, X_0, Z_1, X_1, Y): from

$$P(z_0) \cdot P(x_0 \mid z_0) \cdot P(z_1 \mid x_0, z_0) \cdot P(x_1 \mid z_1, x_0, z_0) \cdot P(y \mid x_1, z_1, x_0, z_0)$$

drop the treatment factors:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_0, z_1} \mathbb{E}[Y \mid x_0, z_0, x_1, z_1]\, P(z_1 \mid x_0, z_0)\, P(z_0).$$

This is the *g-formula*; sequential exchangeability makes each factor estimable. New here: why feedback breaks standard estimation, and weight products, MSMs, SNMs.

11.3.2 The g-formula

Motivation. Standardize over the time-varying confounder instead of conditioning on it.

Idea (two time points, no baseline confounder). Standardize over Z_1 *given past treatment*, not its marginal:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_1} \mathbb{E}[Y \mid X_0 = x_0, Z_1 = z_1, X_1 = x_1]\, P(Z_1 = z_1 \mid X_0 = x_0).$$

Derivation. Temporally,

$$P(y, x_1, z_1, x_0) = P(y \mid x_1, z_1, x_0) \underbrace{P(x_1 \mid z_1, x_0)}_{\text{remove}} P(z_1 \mid x_0) \underbrace{P(x_0)}_{\text{remove}};$$

under $\mathrm{do}(X_0 = x_0, X_1 = x_1)$ the assignment terms drop, leaving outcome $\times$ confounder model. For $K + 1$ time points,

$$\mathbb{E}[Y(\bar{x})] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{X} = \bar{x}, \bar{Z} = \bar{z}] \prod_{k=0}^{K} P(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}),$$

with $\bar{x}_{-1}, \bar{z}_{-1}$ empty, $P(z_0 \mid \bar{x}_{-1}, \bar{z}_{-1}) = P(z_0)$.

11.3.2.1 Iterated conditional expectations (ICE)

The sum over $\bar{z}$ needs densities $P(z_k \mid \text{past})$, hard for high-dimensional Z_k. ICE replaces it by backward nested regressions, the data averaging out one confounder per step:

$$Y \xrightarrow{\text{regress}} \hat{Q}_K \to \cdots \to \hat{Q}_0 \xrightarrow{\text{average}} \hat{\mathbb{E}}[Y(\bar{x})].$$

Two time points. (1) $\hat{Q}_1(z_1) = \hat{\mathbb{E}}[Y \mid x_0, z_1, x_1]$;

(2) $\hat{\mathbb{E}}[Y(x_0, x_1)] = \hat{\mathbb{E}}[\hat{Q}_1(Z_1) \mid X_0 = x_0]$, averaging over the empirical Z_1 in the $X_0 = x_0$ group.

Three time points as variable elimination. Eliminate Z_2, then Z_1:

$$Q_2(z_1, z_2) = \mathbb{E}[Y \mid x_0, z_1, x_1, z_2, x_2], \qquad Q_1(z_1) = \mathbb{E}[Q_2(Z_1, Z_2) \mid x_0, z_1, x_1],$$
$$\mathbb{E}[Y(x_0, x_1, x_2)] = \mathbb{E}[Q_1(Z_1) \mid X_0 = x_0].$$

For Q_1, regress the pseudo-outcome $\tilde{Y}_i^{(2)} = \hat{Q}_2(Z_{1i}, Z_{2i})$ (treatments set to (x_0, x_1, x_2)) on Z_1 among rows with $X_0 = x_0$, $X_1 = x_1$.

General recursion. With $\hat{Q}_{K+1} = Y$, for $k = K, \ldots, 0$:

$$\hat{Q}_k = \hat{\mathbb{E}}[\hat{Q}_{k+1} \mid \bar{X}_k = \bar{x}_k, \bar{Z}_k], \qquad \hat{\mathbb{E}}[Y(\bar{x})] = \hat{\mathbb{E}}[\hat{Q}_0(x_0, Z_0)].$$

ICE never models $P(z_k \mid \text{past})$, so it scales to many time points and high-dimensional Z_k.

11.3.3 IP weighting and marginal structural models

Motivation. Re-weight to a pseudo-population with unconfounded treatment.

Idea. Weight each individual by the inverse probability of their observed treatment given confounders, which "breaks" $Z_k \to X_k$ (U never points into treatment).

One time point: why $1/P(X \mid Z)$. Under exchangeability, consistency and positivity,

$$\mathbb{E}\Big[\frac{\mathbb{1}(X = x)Y}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[\mathbb{1}(X = x)Y \mid Z]}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = \mathbb{E}_Z\big[\mathbb{E}[Y(x) \mid Z]\big] = \mathbb{E}[Y(x)].$$

Up-weighting rare (X, Z) combinations gives a pseudo-population with $X \perp\!\!\!\perp Z$.

Two time points (no baseline confounder), regime (x_0, x_1):

$$W^X(x_0, x_1) = \frac{\mathbb{1}(X_0 = x_0)}{P(x_0)} \cdot \frac{\mathbb{1}(X_1 = x_1)}{P(x_1 \mid x_0, Z_1)},$$

$$\mathbb{E}[W^X(x_0, x_1)Y] = \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] \frac{P(x_0, z_1, x_1)}{P(x_0)P(x_1 \mid x_0, z_1)}$$
$$= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1]P(z_1 \mid x_0) = \mathbb{E}[Y(x_0, x_1)].$$

The factors undo the assignment of X_0 and of X_1 given (X_0, Z_1). With baseline Z_0, use $P(X_0 = x_0 \mid Z_0)$.

The weight product. Divide by $P(X_k \mid \text{past})$ at every k:

$$W^X = \prod_{k=0}^{K} \frac{1}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)} \qquad \text{(non-stabilized)}.$$

Stabilized weights. For binary treatments

$$\mathbb{E}[W^X] = \sum_{\bar{x}, \bar{z}} \frac{P(\bar{x}, \bar{z})}{\prod_k P(x_k \mid \text{past})} = \sum_{\bar{x}, \bar{z}} \prod_k P(z_k \mid \text{past}) = \sum_{\bar{x}} 1 = 2^{K+1},$$

so the pseudo-population is inflated and rare cells give extreme weights. Use the numerator $P(X_k \mid \bar{X}_{k-1})$:

$$SW^X = \prod_{k=0}^{K} \frac{P(X_k \mid \bar{X}_{k-1})}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

It involves no Z, so validity is kept, and $\mathbb{E}[SW^X] = 1$, with much smaller variance.

Proof (one time point). With $SW = P(X)/P(X \mid Z)$,

$$\mathbb{E}\Big[\frac{P(x)}{P(x \mid Z)}\mathbb{1}(X = x)Y\Big] = P(x)\,\mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= P(x)\,\mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = P(x)\,\mathbb{E}[Y(x)].$$

With $Y \equiv 1$, $\mathbb{E}[SW\mathbb{1}(X = x)] = P(x)$, so $\mathbb{E}[SW] = \sum_x P(x) = 1$ and

$$\mathbb{E}[SW\mathbb{1}(X = x)Y]/\mathbb{E}[SW\mathbb{1}(X = x)] = \mathbb{E}[Y(x)].$$

The factor $P(x)$ rescales the $X = x$ arm to its marginal size and cancels in the ratio (and in an MSM).

Proof (general). Let $r_k(x_k \mid \bar{x}_{k-1}) \equiv P(X_k = x_k \mid \bar{X}_{k-1} = \bar{x}_{k-1})$. The observed law is $p(\bar{z}, \bar{x}, y) = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})p(x_k \mid \bar{x}_{k-1}, \bar{z}_k) \cdot p(y \mid \bar{x}, \bar{z})$. For $SW^X = \prod_k r_k(X_k \mid \bar{X}_{k-1})/p(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$,

$$p_{SW}(\bar{z}, \bar{x}, y) = p(\bar{z}, \bar{x}, y)\, SW^X = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r_k(x_k \mid \bar{x}_{k-1}) \cdot p(y \mid \bar{x}, \bar{z}),$$

a proper density (so $\mathbb{E}[SW^X] = 1$) without $Z_k \to X_k$ arrows. For static $\bar{x}$ let $r(\bar{x}) = \prod_k r_k(x_k \mid \bar{x}_{k-1})$. By the g-formula

$$\mathbb{E}\big[SW^X\mathbb{1}(\bar{X} = \bar{x})Y\big] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{x}, \bar{z}] \prod_k p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r(\bar{x}) = r(\bar{x})\,\mathbb{E}[Y(\bar{x})],$$

and $\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})] = r(\bar{x})$, so $\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})Y]/\mathbb{E}[SW^X\mathbb{1}(\bar{X} = \bar{x})] = \mathbb{E}[Y(\bar{x})]$.

11.3.3.1 Marginal structural models

An MSM models the counterfactual mean as a function of strategy, e.g.

$$\mathbb{E}[Y(x_0, x_1)] = \beta_0 + \beta_1 x_0 + \beta_2 x_1 + \beta_3 x_0 x_1,$$

fitted by weighted regression in the IP-weighted pseudo-population. Variants: saturated (one parameter per strategy); dose–response $\mathbb{E}[Y(\bar{x})] = \beta_0 + \beta_1 \mathrm{cum}(\bar{x})$, $\mathrm{cum}(\bar{x}) = \sum_k x_k$; conditional MSMs with baseline covariates.

11.3.4 G-estimation of structural nested models

Motivation. Model and remove the history-specific effect of each decision.

The blip. An SNMM models the history-specific effect of each decision, the *blip*, with the future at reference (0):

$$\gamma_1(x_0, z_1) = \mathbb{E}\big[Y(x_0, 1) - Y(x_0, 0) \mid X_0 = x_0, Z_1 = z_1\big] \text{ (time-1 blip, given history)},$$

$$\gamma_0(x_0) = \mathbb{E}\big[Y(x_0, 0) - Y(0, 0)\big] \text{ (time-0 blip)}.$$

From 000 to 111, $\mathbb{E}[Y(\bar{x})]$ climbs $000 \to 100 \to 110 \to 111$ by $+\gamma_0, +\gamma_1, +\gamma_2$; the MSM models the whole climb, the SNMM the steps.

G-estimation: find the blip by removing it. $Y_{\text{removed}} = Y - \psi_1 X_1$ rebuilds the outcome without the late treatment; choose ψ_1 so that Y_{removed} is balanced across X_1 given history (an estimating equation). IPW models treatment; SNMM models the effect; both recover $\mathbb{E}[Y(\bar{x})]$.

In practice: one time point. With blip $\gamma(z;\psi) = \psi x$, $H(\psi) = Y - \psi X$. At the true ψ, $H(\psi) = Y(0) \perp\!\!\!\perp X \mid Z$, so $H(\psi)$ is uncorrelated with X given Z. With $\hat{e}(Z) = \hat{P}(X = 1 \mid Z)$,

$$U(\psi) = \sum_i (X_i - \hat{e}(Z_i)) \underbrace{(Y_i - \psi X_i)}_{H_i(\psi)} = 0 \quad \Longrightarrow \quad \hat{\psi} = \frac{\sum_i (X_i - \hat{e}_i) Y_i}{\sum_i (X_i - \hat{e}_i) X_i},$$

an IV-type ratio with instrument $X - \hat{e}$. Recipe:

(1) fit $\hat{e}(Z)$ (e.g. logistic);

(2) form residuals $X_i - \hat{e}_i$;

(3) plug in. Consistent if $\hat{e}$ is correct; an outcome term $\mathbb{E}[H(\psi) \mid Z]$ makes it doubly robust. With $\gamma(z;\psi) = (\psi_0 + \psi_1 z)x$, the instrument is $(1, z)(X - \hat{e})$.

Two time points. $X_0 \to Z_1 \to X_1 \to Y$, blips $\gamma_1 = \psi_1 x_1$, $\gamma_0 = \psi_0 x_0$; latest blip first.

1. $H_1(\psi_1) = Y - \psi_1 X_1$ is the candidate $Y(x_0, 0)$, with $Y(x_0, 0) \perp\!\!\!\perp X_1 \mid (X_0, Z_1)$. With $\hat{e}_1 = \hat{P}(X_1 = 1 \mid X_0, Z_1)$, $\sum_i (X_{1i} - \hat{e}_{1i}) H_{1i}(\psi_1) = 0$ gives
$$\hat{\psi}_1 = \frac{\sum_i (X_{1i} - \hat{e}_{1i}) Y_i}{\sum_i (X_{1i} - \hat{e}_{1i}) X_{1i}}.$$
2. $H_0 = H_1(\hat{\psi}_1) - \psi_0 X_0 = Y - \hat{\psi}_1 X_1 - \psi_0 X_0$ is the candidate $Y(0, 0)$. With $\hat{e}_0 = \hat{P}(X_0 = 1 \mid Z_0)$,
$$\hat{\psi}_0 = \frac{\sum_i (X_{0i} - \hat{e}_{0i}) H_{1i}(\hat{\psi}_1)}{\sum_i (X_{0i} - \hat{e}_{0i}) X_{0i}}.$$

Telescoping, $\mathbb{E}[Y(1,1)] - \mathbb{E}[Y(0,0)] = \hat{\psi}_0 + \hat{\psi}_1$; outcome terms $\mathbb{E}[H_k \mid \text{history}]$ make each step doubly robust.

11.3.5 Doubly robust estimation

Motivation. Combine outcome and treatment models for two chances at consistency.

The g-formula needs a correct $\mathbb{E}[Y \mid \bar{X}, \bar{Z}]$, IPW correct $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$; AIPW is consistent if either is correct.

Time-fixed review. With $\hat{m}(x, z) = \hat{\mathbb{E}}[Y \mid X = x, Z = z]$ and $\hat{\pi}(z) = \hat{P}(X = 1 \mid Z = z)$,

$$\hat{\tau}_{DR} = \frac{1}{n} \sum_{i=1}^{n} \left[\hat{m}(1, Z_i) - \hat{m}(0, Z_i) + \frac{X_i \{Y_i - \hat{m}(1, Z_i)\}}{\hat{\pi}(Z_i)} - \frac{(1 - X_i)\{Y_i - \hat{m}(0, Z_i)\}}{1 - \hat{\pi}(Z_i)} \right].$$

If $\hat{m}$ is correct the residual terms have mean zero; if $\hat{\pi}$ is correct they remove the bias of $\hat{m}$.

Time-varying extension. Fit $\hat{\pi}_k$ at each time and outcome models backward (ICE-style); consistent if at each time either model is correct (Bang and Robins 2005). Modern variants: TMLE, sequential regression with cross-fitting.

11.3.6 Practical considerations

Motivation. Dynamic strategies and positivity.

Dynamic strategies. For $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, $g_k(\cdot) \in \{0, 1\}$. G-formula: simulate forward, drawing Z_k and setting $X_k = g_k(\cdot)$. IPW: weight those who followed g,

$$W^g = \prod_{k=0}^{K} \frac{\mathbb{1}(X_k = g_k(\bar{X}_{k-1}, Z_k))}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

G-estimation: SNMMs handle dynamic regimes via effect modification by Z_k.

Positivity violations. K binary times give 2^K histories, so cells are sparse, weights extreme, estimates unstable. Remedies: stabilized or truncated weights, parametric models.

11.3.7 Comparison of g-methods

Motivation. What each method models, targets and risks.

	G-formula	IPW / MSM	G-estimation / SNM
Key model	outcome + confounder dist.	treatment assignment	treatment + blip function
Target	$\mathbb{E}[Y(\bar{x})]$ (marginal)	$\mathbb{E}[Y(\bar{x})]$ (marginal)	conditional blip effects
Effect modification by Z_k	outcome-model interactions	not directly	directly modeled
Extreme weights?	no	yes	no
G-null paradox?	yes (parametric)	no	no
DR version?	yes	yes (AIPW)	yes

All are consistent under correct specification and the same conditions. (G-null paradox: parametric outcome and confounder models cannot all be correct under the null, so the parametric g-formula may reject a true null.)

Estimand, model, estimator.

Method	Estimand (what)	Model (what must be right)	Estimator (how)
g-formula	$\mathbb{E}[Y(\bar{x})]$	outcome (+ confounders)	standardize / ICE
IPW	$\mathbb{E}[Y(\bar{x})]$	treatment	weighting
MSM	$\mathbb{E}[Y(\bar{x})]$, marginal	*marginal form* + treatment	weighted regression
SNMM / g-est.	blip γ_k (history-specific)	*blip form* + treatment	g-estimation (balance)
AIPW / DR	$\mathbb{E}[Y(\bar{x})]$	outcome or treatment	augmented IPW

Structural models (italic) restrict the estimand; *nuisance* models describe observed data. MSMs are fitted by IPW, SNMMs by g-estimation; only the SNMM targets the history-specific blip, enabling effect modification and optimal dynamic regimes.

The big picture. The g-formula standardizes over Z, IPW/MSM removes $Z \to X$ arrows, g-estimation subtracts blips backward, AIPW combines both models.

Key takeaways.

1. Time-varying treatments need multiple time points and treatment strategies.
2. With confounders affected by prior treatment, regression, stratification and matching fail.
3. G-methods work under sequential exchangeability, consistency and positivity.
4. Doubly robust methods protect against misspecification.
5. Choose by problem structure, available models and positivity.